Box-counting dimension

Let F be a nonempty bounded subset of $\mathbb{R}^n$. For $\delta > 0$, let $N_\delta(F)$ be the least number of sets of diameter at most δ which can cover F.

We define the lower box-counting dimension, $\underline{\dim}_B F$ and the upper box-countind dimension, $\overline{\dim}_B F$ as follows:

$$\underline{\dim}_B F = \liminf_{\delta \to 0^+} \frac{\ln N_\delta(F)}{-\ln \delta}$$

and $$\overline{\dim}_B F = \limsup_{\delta \to 0^+} \frac{\ln N_\delta(F)}{-\ln \delta}.$$

Clearly, $\underline{\dim}_B F \le \overline{\dim}_B F$.

If $\underline{\dim}_B F = \overline{\dim}_B F$, we call the common value as the box-counting dimension of F, denoted by $\dim_B F$.

i.e. $$\dim_B F = \lim_{\delta \to 0^+} \frac{\ln N_\delta(F)}{-\ln \delta}.$$

If $s = \dim_B F$, $N_\delta(F) \simeq c\,\delta^{-s}$ for small δ.

In fact,

$$\lim_{\delta \to 0^+} \left(N_\delta(F)\,\delta^s\right) = +\infty \quad \text{if } s < \dim_B F$$

$$\lim_{\delta \to 0^+} \left(N_\delta(F)\,\delta^s\right) = 0 \quad \text{if } s > \dim_B F.$$

Theorem: In the definition of $\underline{\dim}_B F$ & $\overline{\dim}_B F$, we can replace $N_\delta(F)$ by any one of the following:

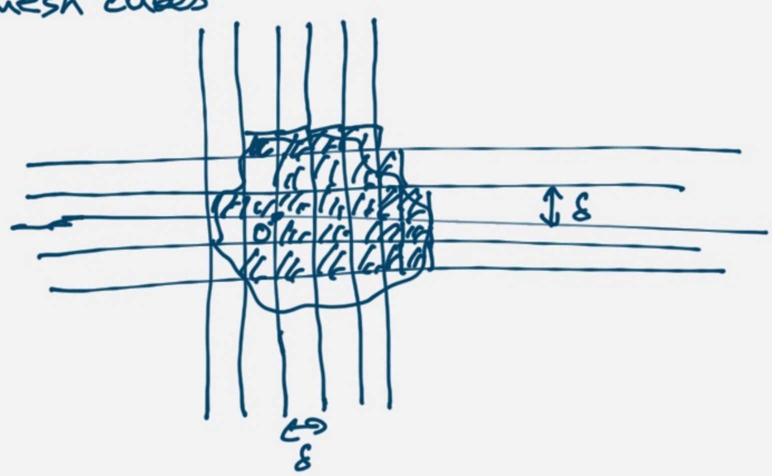
(i) the least number of closed balls of radius δ that cover F.

(ii) the least number of cubes of side length δ that cover F.

(iii) the number $\underline{\delta\text{-mesh}}$ cubes that intersect F.

(iv) the largest number of disjoint balls of radius δ with centres in F.

"δ-mesh cubes"



Example: Let F be the Cantor set.
Then $\dim_B F = \frac{\ln 2}{\ln 3}$.

Proof: We'll show that $\overline{\dim}_B F \leq \frac{\ln 2}{\ln 3}$
and $\underline{\dim}_B F \geq \frac{\ln 2}{\ln 3}$ and then we are done.

Let $0<\delta<1$ and let $k\in\mathbb{N}$ s.t.
$3^{-k} < \delta \leq 3^{-k+1}$.

Then the 2^k intervals of length 3^{-k} at the kth level of the Cantor set provides a δ-cover of F.

$\Rightarrow N_\delta(F) \leq 2^k \Rightarrow \ln N_\delta(F) \leq k\ln 2$

Also, $\delta \leq 3^{-k+1} \Rightarrow -\ln\delta \geq (k-1)\ln 3$

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$$\therefore \frac{\ln N_\delta(F)}{-\ln\delta} \leq \frac{k\ln 2}{(k-1)\ln 3}$$

$$\Rightarrow \overline{\dim}_B F = \overline{\lim_{\delta\to 0^+}} \frac{\ln N_\delta(F)}{-\ln\delta} \leq \lim_{k\to\infty}\left(\frac{k}{k-1}\right)\frac{\ln 2}{\ln 3} = \frac{\ln 2}{\ln 3}.$$

Also, if $3^{-k-1} \leq \delta < 3^{-k}$, then any set of diameter $<\delta$ can intersect at most one of the 2^k intervals of length 3^{-k}.

$\therefore N_\delta(F) \geq 2^k$.

$$\therefore \underline{\dim}_B F = \underline{\lim_{\delta\to 0^+}} \frac{\ln N_\delta(F)}{-\ln\delta} \geq \lim_{k\to\infty}\frac{\ln(2^k)}{-\ln(3^{-k-1})} = \lim_{k\to\infty}\frac{k\ln 2}{(k+1)\ln 3} = \frac{\ln 2}{\ln 3}.$$

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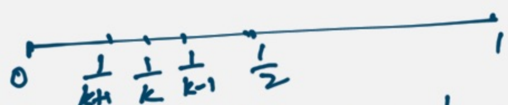
<u>Exercise</u>: Prove that the box-counting dimension of the Sierpinski triangle is $\frac{\ln 3}{\ln 2}$.

<u>Example</u>: Let $F = \{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \cdots\}$.

Then $\dim_B F =$

<u>Proof</u>: Let $0 < \delta < \frac{1}{2}$



Let $k \in \mathbb{N}$ s.t. $\frac{1}{k} - \frac{1}{k+1} \leq \delta < \frac{1}{k-1} - \frac{1}{k}$

i.e., $\frac{1}{k(k+1)} \leq \delta < \frac{1}{k(k-1)}$

If $|U| \leq \delta$, then U can cover at most one of the points $1, \frac{1}{2}, \cdots, \frac{1}{k}$ because the least distance between any two of these points is $\frac{1}{k-1} - \frac{1}{k} > \delta$.

$\therefore \quad N_\delta(F) \geq k \Rightarrow \ln N_\delta(F) \geq \ln k$

Also, $\delta \geq \frac{1}{k(k+1)} \Rightarrow -\ln \delta \leq \ln(k(k+1))$

$\therefore \quad \frac{\ln N_\delta(F)}{-\ln \delta} \geq \frac{\ln k}{\ln k + \ln(k+1)} \to \frac{1}{2}$ as $k \to \infty$.

$\Rightarrow \boxed{\underline{\dim}_B F \geq \frac{1}{2}}$

On the other hand, if $\frac{1}{k(k+1)} \leq \delta < \frac{1}{k(k-1)}$,

then $(k+1)$ intervals of length δ can cover the interval $[0, \frac{1}{k}]$.

$\therefore \quad N_\delta(F) \leq (k+1)+(k-1) = 2k$

$$\Rightarrow \quad \frac{\ln N_\delta(F)}{-\ln \delta} \leq \frac{\ln(2k)}{\ln(k(k-1))} = \frac{\ln 2+\ln k}{\ln k+\ln(k-1)}$$

$$\rightarrow \frac{1}{2} \text{ as } k \to \infty$$

$\therefore \quad \boxed{\overline{\dim}_B F \leq \frac{1}{2}}$

$\therefore \quad \boxed{\dim_B F = \frac{1}{2}}$ for $F = \{\frac{1}{n} : n \in \mathbb{N}\}$

Exercise: Find the box-counting dimension for

(i) $F = \{\frac{1}{n^2} : n \in \mathbb{N}\}$

(ii) $F = \{\frac{1}{n^3} : n \in \mathbb{N}\}$

(iii) $F = \{\frac{1}{2^n} : n \in \mathbb{N}\}$.

Some properties of box-counting dimension

(i) Monotonicity: If $E \subseteq F$, then $\underline{\dim}_B E \leq \underline{\dim}_B F$ and $\overline{\dim}_B E \leq \overline{\dim}_B F$.

(ii) Range of values: For any nonempty bounded subset of $\mathbb{R}^n$,

$$0 \leq \underline{\dim}_B F \leq \overline{\dim}_B F \leq n.$$

(iii) Open sets: For any nonempty bounded open subset U of $\mathbb{R}^n$, $\dim_B U = n$.

(Use monotonicity with cubes contained inside and containing U).

(iv) $\overline{\dim}_B$ is finitely stable:

$$\overline{\dim}_B(E\cup F) = \max\{\overline{\dim}_B E, \overline{\dim}_B F\}.$$

Pf: $N_\delta(E\cup F) \le N_\delta(E)+N_\delta(F) \le 2\max\{N_\delta(E), N_\delta(F)\}$

$$\therefore \overline{\dim}_B(E\cup F) = \limsup_{\delta\to 0^+} \frac{\ln N_\delta(E\cup F)}{-\ln\delta}$$

$$\le \limsup_{\delta\to 0^+} \frac{\ln 2 + \max\{\ln N_\delta(E), \ln N_\delta(F)\}}{-\ln\delta}$$

$$= \max\{\overline{\dim}_B E, \overline{\dim}_B F\}$$